

comodate an additional switch. Also, use a heat-sink with LM317 to handle the desired amount of power.

Assemble the circuit on a small general-purpose PCB and enclose in a

suitable case. Fit the entire PCB inside a glue stick tube as shown in Fig. 2. Affix the female and male connectors on the opposite ends and place the ammeter in between the stick tube. You

can directly read the output voltage on the ammeter after due calibration. ●

Note. You can use a suitable VU meter instead of 0-100µA ammeter and calibrate accordingly.

Driveway Alarm

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You can install this system in the driveway itself to detect intrusion by a vehicle, person or large animal. It sounds an alarm when any intrusion takes place.

the pointer and connect to the battery contact spring. Hold the laser on/off switch closed with a small cable tie. The other clip can then be connected to the body of the laser torch through R1. As laser pointers are not designed for this type of application, select the

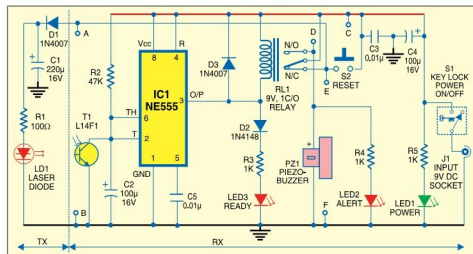


Fig. 1: Circuit of driveway alarm

As shown in Fig. 1, the system comprises two modules: transmitter and receiver. The transmitter module (TX) is wired around a handheld laser pointer device (LD1). The laser pointer is powered by the 9V DC supply available from the receiver module (RX). Installation of the transmitter and receiver modules is shown in Fig. 2. Resistor R1 limits the current flowing through the laser pointer (LD1).

Laser pointers are not designed to be taken apart. So the following method might be used to connect the external power supply to the laser diode without damaging the pointer itself: Take two short lengths of flexible wires with small metal clips soldered at their one end. Insert one of the metal clips inside the body of

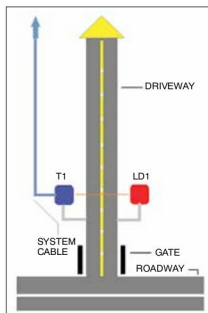


Fig. 2: Installation plan

value of R1 carefully.

The receiver module (RX) is a compact light-controlled security switch, powered by an external power adaptor (230V AC 50Hz/9V DC 250mA).

A key-lock switch (S1) controls the power supply to the receiver module. There are two LEDs on the front panel for power (LED1) and visitor alert (LED2).

Reset switch S2 on the front panel can be used to clear the alert signal after detection of a driveway activity. LED3 indicates that the system is ready.

Working of the receiver module is very simple. Initially, when power is applied to the system, it is in standby mode. LED1 indicates the presence of power and LED3 lights up to indicate that the laser beam from the transmitter module is falling on the photosensor (T1) of the receiver module. When the laser beam is interrupted by a person, animal or object, the photosensor energises the rest of the circuit and the piezobuzzer sounds.

Assemble the circuit on a general-purpose PCB. Fabricate the transmitter, receiver and the control panel (an extension of the receiver module) as three independent units, sharing a common power supply. House these units in suitable waterproof plastic enclosures and interconnect using high-quality cables.

Install the transmitter and receiver modules far enough from the driveway edge so that the vehicles are unlikely to hit these. Place the transmitter module on one side of the driveway, approximately one metre

